Tip: the journey bar is clickable - jump back to any finished stage, or between your tests, anytime.

RESULTS

Results

01 · MANCOVA

MANOVA with covariate control

Table 1. Multivariate tests (covariate-adjusted)

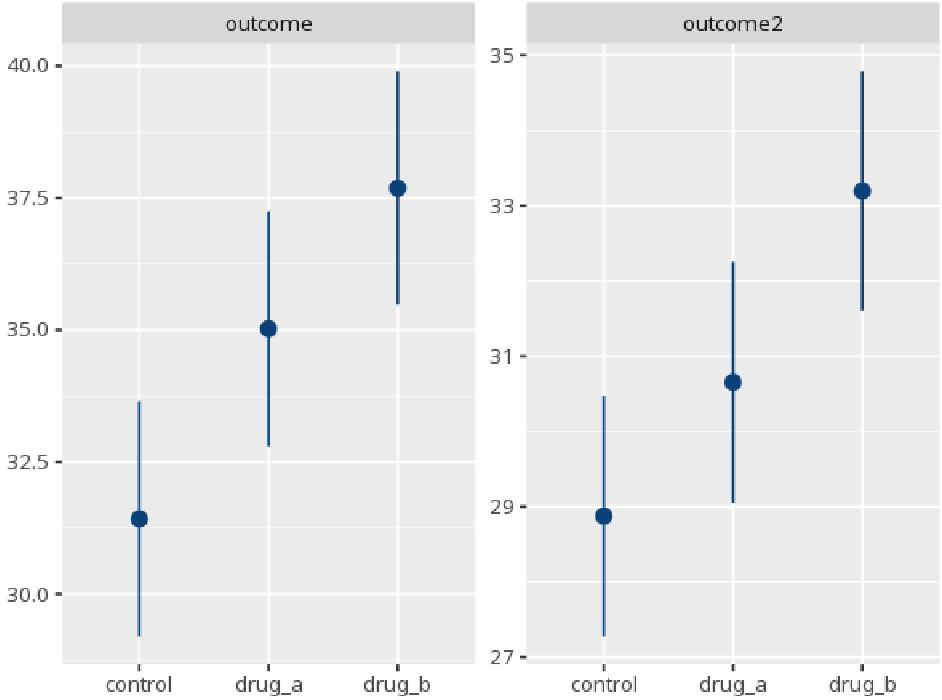
Effect	Pillai / Wilks	approx F	df1	df2	p	partial η² [95% CI]
baseline	0.55	33.87	2	55	<.001	0.55 [0.40, 1.00]
group	0.37	6.30	4	112	<.001	0.18 [0.07, 1.00]

Table 2. Adjusted univariate follow-ups

DV	F	df1	df2	p	partial η² [95% CI]
outcome	8.07	2	56	<.001	0.22 [0.07, 1.00]
outcome2	7.47	2	56	.001	0.21 [0.06, 1.00]

assumption checks include homogeneity of covariance matrices (Box's M) and homogeneity of regression slopes for each covariate. (Box's M $\chi^2(6)$=11.44, p=.076 · slopes p(baseline × group)=.421) — covariance matrices look homogeneous — the regression slopes look homogeneous across groups

Figure. Adjusted means per outcome



Understanding the numbers

Effect. Names which factor (or interaction) the covariate-adjusted multivariate test is evaluating. Here, the effect tested is group.

Pillai's V / Wilks' Λ. The chosen multivariate test statistic summarizing how much the groups differ across all outcomes jointly, after controlling for the covariate(s). Here, Pillai's V = 0.37.

approx F. An F-value approximating the covariate-adjusted multivariate test's significance; the follow-up table reports the same kind of F computed per single outcome. Here, approx $F(4, 112) = 6.30$.

df1. The numerator degrees of freedom for the multivariate (or follow-up) test. Here, df1 = 4.

df2. The denominator degrees of freedom for the multivariate (or follow-up) test. Here, df2 = 112.

p. The probability of a covariate-adjusted multivariate difference this large if the groups truly had identical outcome profiles. Here, $p < .001$ for the multivariate test - below the $\alpha = 0.05$ threshold.

partial η^2 (multivariate). The effect size for the covariate-adjusted multivariate test itself, converted directly from its approx F and degrees of freedom (there is no per-cell sum-of-squares breakdown for a multivariate statistic). Here, partial $\eta^2 = .18$ [.07, 1.00] for the multivariate test.

DV. Names which single outcome that follow-up row's covariate-adjusted univariate ANOVA tests. Here, the follow-up table above reports one row for each of the 2 outcomes.

partial η^2. The proportion of that single outcome's variance explained by the factor after controlling for the covariate(s) - adjust these follow-up p-values for the number of DVs before calling any significant. Here, each of the 2 follow-up ANOVAs above reports its own partial η^2 for that outcome.

How to read this test

Like MANOVA, but group differences on the set of outcomes are assessed after controlling for one or more covariates. Interpret the univariate follow-ups only if the multivariate p is significant, and adjust them for the number of DVs (e.g. Bonferroni) to control familywise error. Your significance threshold (α) is 0.05.

APA template: A MANCOVA gave a covariate-adjusted group effect, Pillai's V=.37, $F(4,112)=6.30$, $p < .001$, partial η^2=.18 [.07, 1.00].

Statistical basis: Pillai's trace multivariate test (headline statistic, covariate-adjusted) (Pillai, K. C. S. (1955). "Some new test criteria in multivariate analysis." Annals of Mathematical Statistics, 26(1), 117-121.); Wilks' Lambda multivariate test (covariate-adjusted) (Wilks, S. S. (1932). "Certain generalized distributions in multivariate analysis." Biometrika, 24(3/4), 471-494.); Box's M test of homogeneity of covariance matrices (Box, G. E. P. (1949). "A general distribution theory for a class of likelihood criteria." Biometrika, 36(3-4), 317-346.); Multivariate effect size (partial η^2 from the multivariate approx. F) (Ben-Shachar MS, Lüdecke D, Makowski D (2020). "effectsize: Estimation of Effect Size Indices and Standardized Parameters." Journal of Open Source Software, 5(56), 2815.); Estimated marginal (covariate-adjusted) means (Lenth RV (2024). emmeans: Estimated Marginal Means, aka Least-Squares Means. R package.). Full references in the exported CITATIONS.txt.